C4.4 State change descriptions

The cache line state changes associated with a transaction are defined in the following sections:

- *Read transactions* on page C4-191
- *Clean transactions* on page C4-197
- *Make transactions* on page C4-200
- *Write transactions* on page C4-202
- *Evict transactions* on page C4-206.

For each transaction, the starting state for the transaction and the three possible end state groups are given. The three possible end state groups are:

- The expected end states, which are also the end states that this specification recommends.
- The full list of legal end states for a cached master that supports an external snoop filter. This set of end states takes into account that:
  - a cache line in UniqueClean state can always be held in SharedClean state
  - a cache line in UniqueDirty state can always be held in SharedDirty state.
- The full list of legal end states for a cached master that does not support an external snoop filter. This full list of legal end states includes the legal end states for external snoop filter support, and takes into account that a cache line in UniqueClean state, or SharedClean state, can be in the Invalid state.

Some transactions have two tables provided. The first table shows the expected starting states for which the transaction is issued. The second table shows the other permitted starting states for which the transaction is not normally issued. For example, a ReadShared transaction with a Valid starting state. Typically, the transaction and starting state combinations in the second table are associated with a speculative read where the master issues a transaction before it has determined the state of the cache line in its local cache.

Any state not shown as a starting state in the tables is not a legal starting state.

The starting state is defined as the cache line state just before the transaction response is received by the initiating master. If the initiating master receives a snoop transaction to the same cache line between issuing a transaction and receiving the associated response, then the cache line state changes required by the snoop transaction must be applied first. See Chapter C5 *Snoop Transactions*.

The following abbreviations are used for the cache line states:

UC UniqueClean.
UD UniqueDirty.
SC SharedClean.
SD SharedDirty.
I Invalid.

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